



Developer Burnout, Layoff Risk & Career Stability Analytics

An analysis of workforce data from **5,000 software professionals** — identifying burnout drivers, layoff risk factors, AI-related fears, and career stability patterns using Python, PostgreSQL, and Power BI.

The Business Problem

Rising Burnout

Employee burnout and mental health concerns are increasing across tech organizations.

Layoff Anxiety

Workforce instability and reduced job security confidence are driving voluntary attrition.

AI Disruption Fears

Growing concerns around AI replacing technical roles are undermining career confidence.

- ❗ HR leaders need data-driven insights to identify at-risk segments and implement proactive workforce strategies.



Dataset & Analysis Framework



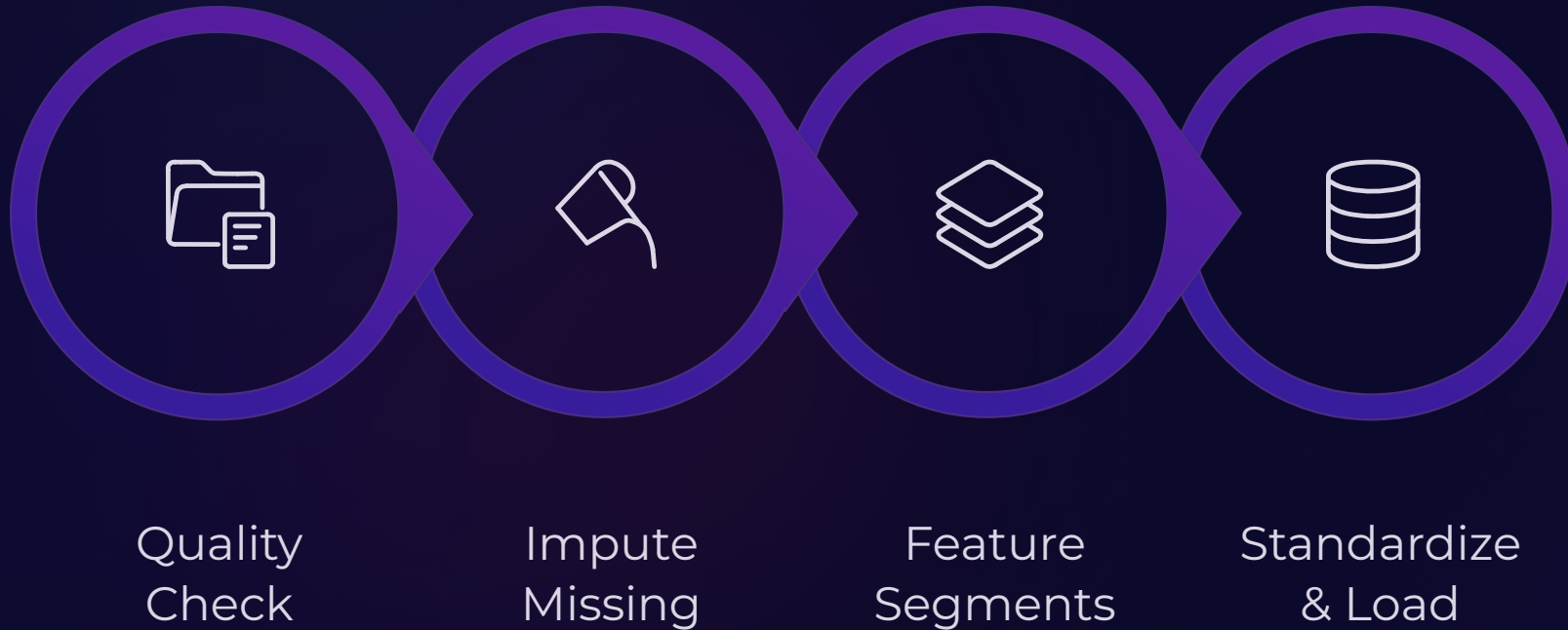
5,000 Software Professionals

Data covers five key categories:

- Demographics: age, gender, education, location
- Employment: role, company type/size, experience, salary
- Mental Health: burnout, stress, anxiety, sleep, work-life balance
- Career Stability: job security, layoff anxiety, promotion, job-switching
- AI Readiness: AI tool usage, replacement fear, upskilling hours

Tools: **Python, PostgreSQL, Power BI** — 34 business-focused analytical questions answered.

Data Preparation & Feature Engineering



Key fixes included correcting Excel-induced date format errors in `company_size` (e.g., "01-Oct" → "1-10"), imputing missing values using medians for numerical features and mode for categorical, and segmenting employees into 5 experience groups: Fresher, Early Career, Mid-Level, Senior, and Veteran.

Executive Overview

5,000

Total Employees

4.46

Avg Burnout Score

4.08

Avg Anxiety Score

2.72%

High Layoff Risk

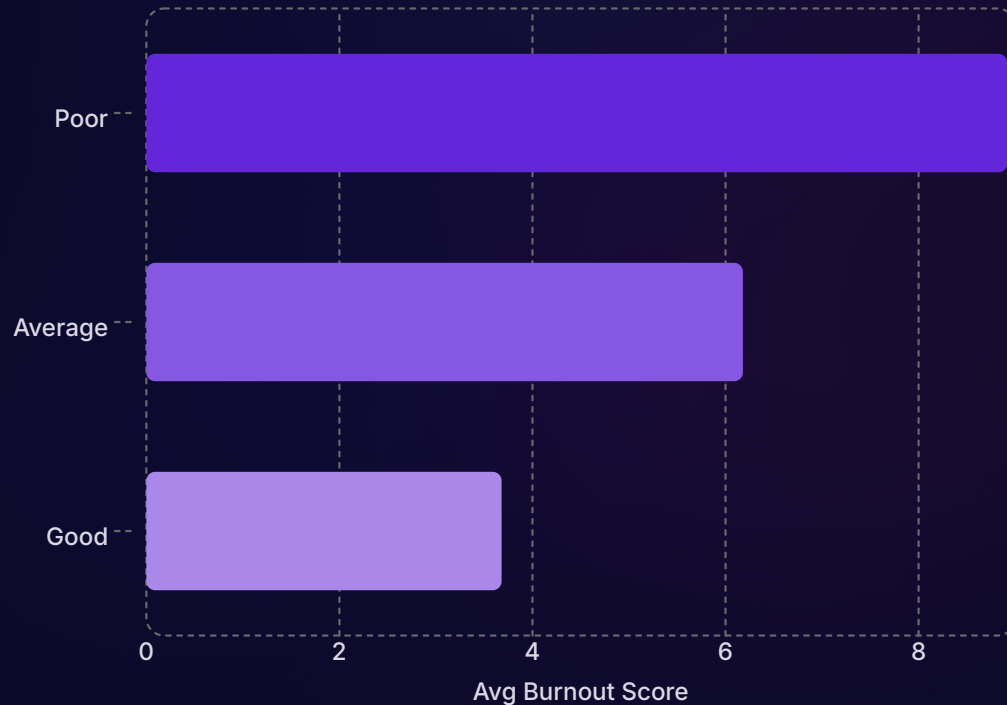
7.22

Avg Job Security

Although only 2.72% fall into high layoff risk, **~40% of employees** are in medium or high burnout categories — revealing broader workforce well-being concerns. Bengaluru leads in employee concentration (1,164), and average salary ranges from 21.35 LPA (Fresher) to 56.39 LPA (Veteran).

Burnout & Mental Health Analysis

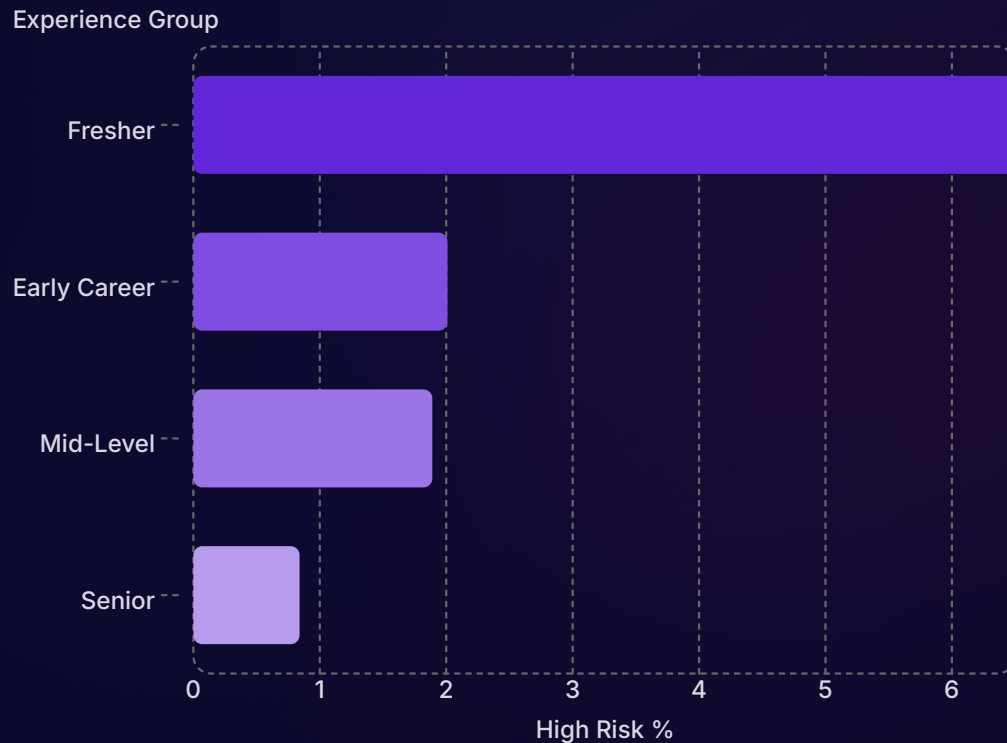
Work-Life Balance



Key Burnout Insights

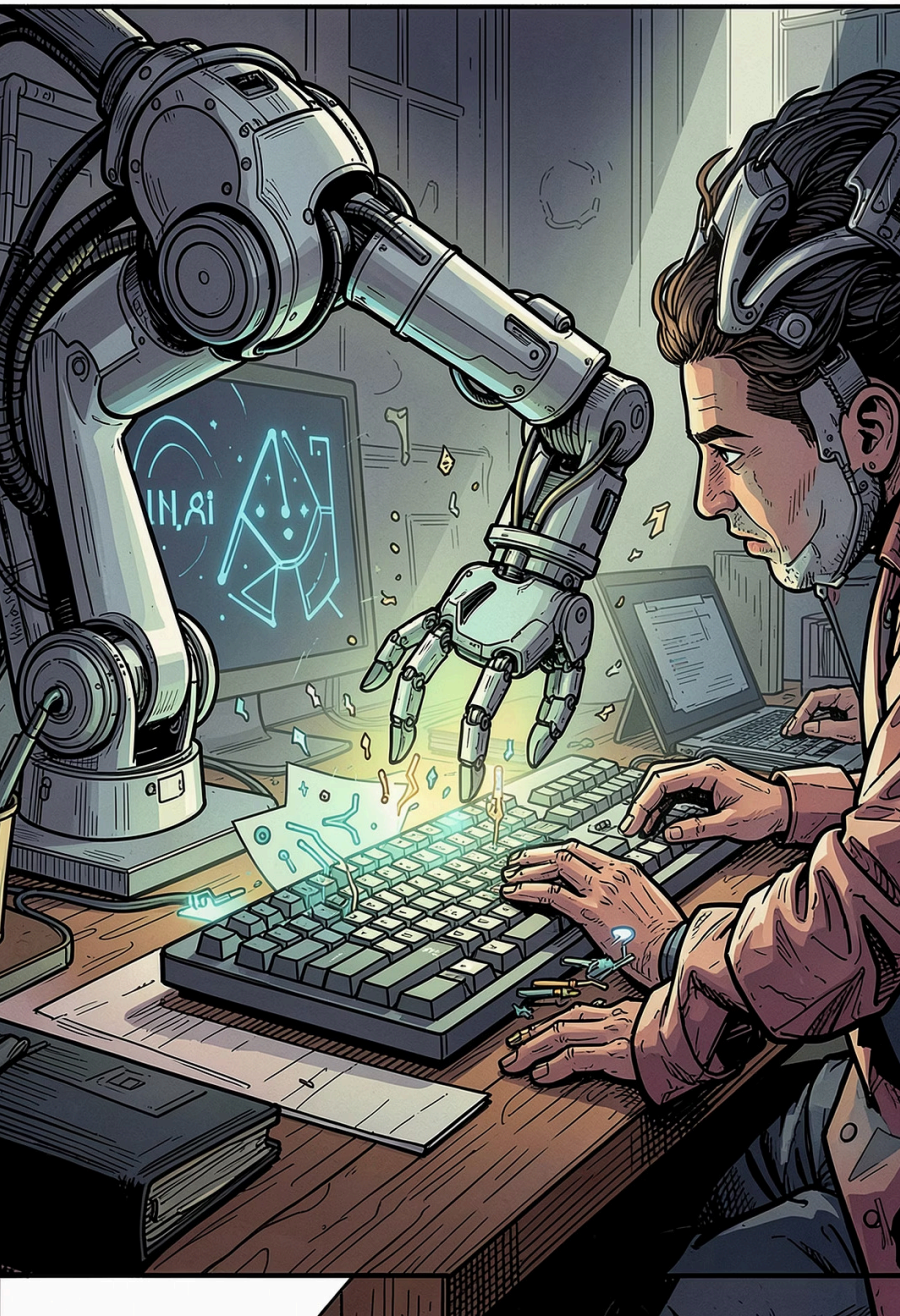
- **Work-life balance** is the strongest predictor — poor balance yields burnout levels 2.4× higher than good balance
- Employees with fewer sleep hours report significantly higher burnout
- ~40% of employees fall into medium or high burnout risk categories
- Early Career professionals show the highest stress and anxiety scores

Layoff Risk Analysis



Who's Most at Risk?

- **Freshers** face 6.5% high layoff risk — significantly more vulnerable than experienced professionals
- **Startups** (3.85%) and **Freelance** (3.41%) show highest company-type risk
- **Small companies** (1–10 employees) carry 4.12% high risk vs. 1.65% at large firms
- High-risk employees report substantially lower job security confidence (5.00 vs 7.29)



AI Fear & Career Stability

AI Fear by Experience

Freshers report the highest AI replacement fear (6.63), declining steadily with experience. Veterans score lowest at 3.87.

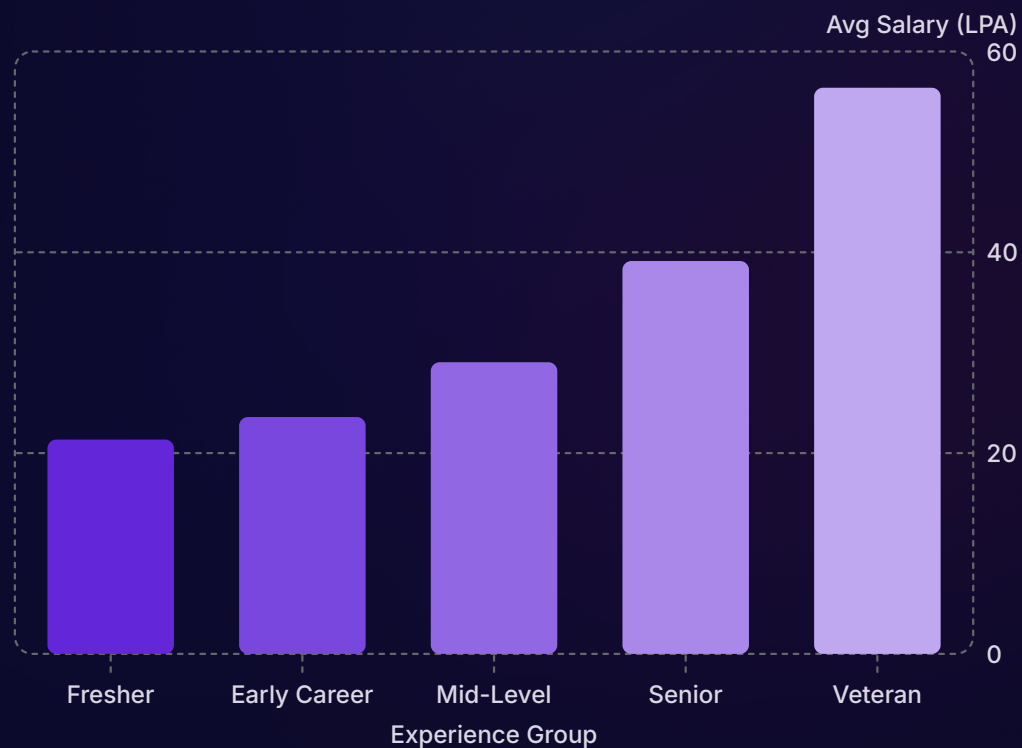
AI Fear → Job Security

Job security drops from 8.62 (low fear) to 6.11 (very high fear). Higher AI fear drives LinkedIn activity and job applications.

Job Switching Intent

Early-career professionals show the highest intention to switch jobs — a critical retention risk for organizations.

Career Growth & Salary Progression



Career Progression Insights

- Salary grows consistently with experience – Veterans earn **2.6× more** than Freshers
- Promoted employees report **lower burnout** (4.43 vs 4.48) and reduced layoff risk
- Early-career professionals show the highest job-switching intention – priority for retention programs
- Financial preparedness (emergency savings) reduces layoff anxiety



Key Findings & Recommendations



Work-Life Balance

Implement flexible scheduling and wellness initiatives — the #1 burnout driver.



Career Growth

Establish transparent promotion pathways to improve retention and stability.



AI Upskilling

Provide AI literacy programs to reduce replacement fears, especially for early-career staff.



Mental Health

Expand counseling access and track burnout/anxiety metrics proactively.

✔ Organizations that proactively address these areas can improve satisfaction, reduce turnover, and strengthen long-term workforce resilience.